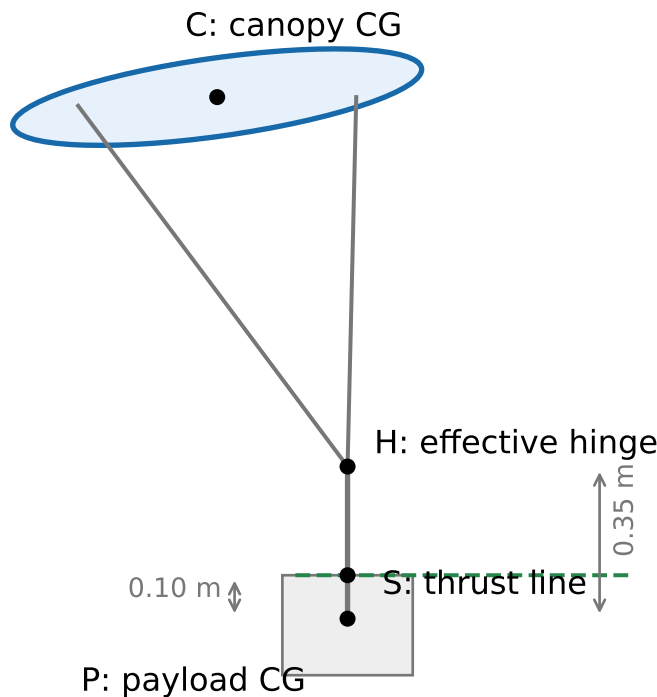


Powered paraglider: steady, straight, level cruise

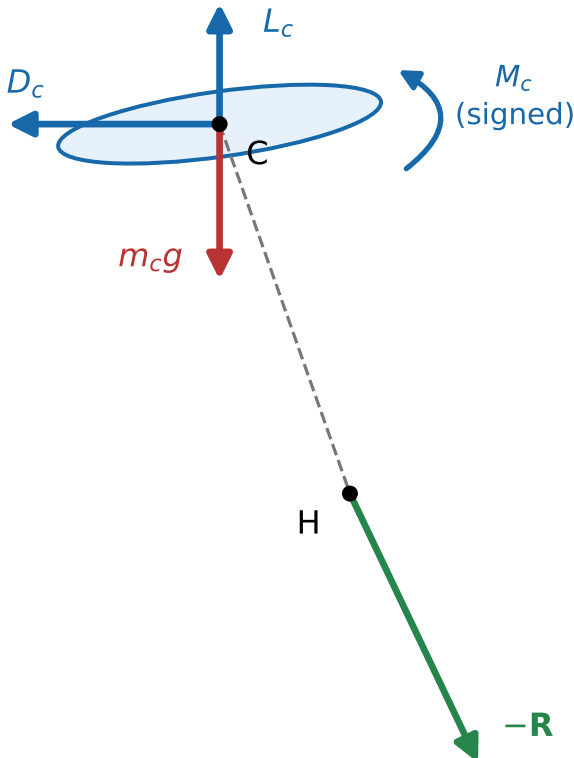
Side view • forward → • forces schematic, arrows not to scale • roll/yaw omitted

Cruise geometry



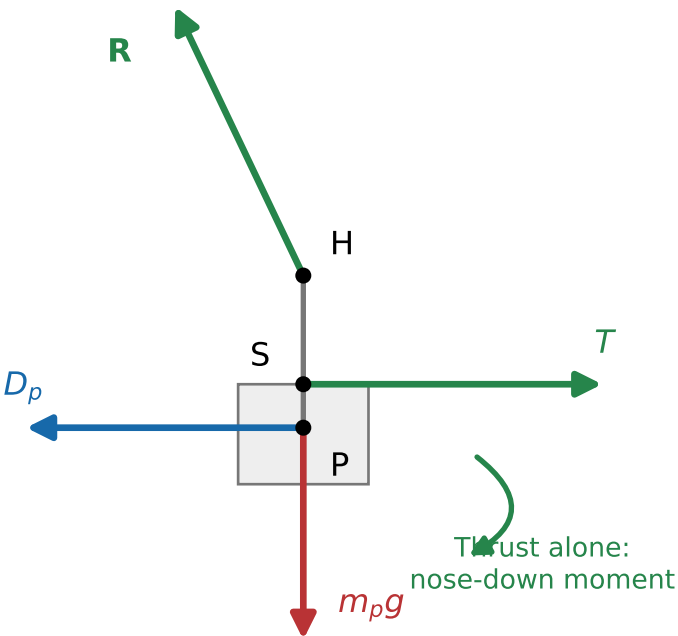
Thrust is below H, but above P.
Offsets shown are provisional sim values.

Canopy free body



Aerodynamic force at C plus pitching moment.
H receives the payload reaction.

Payload free body



$\mathbf{R}$ = canopy force on payload at H; canopy receives $-\mathbf{R}$. This is a hinge reaction, not necessarily a vertical line force.

$$L_c = (m_p + m_c)g \quad T = D_c + D_p \quad \sum M_P = \sum M_C = 0 \text{ at steady cruise.}$$

Payload: thrust above its CG pitches it nose-down; the suspension reaction supplies the balancing moment.
Free pin: no steady joint couple; mechanical joint damping acts only when canopy and payload have relative pitch rate.